The Cosmic Neutrino Background from Recognition Science

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Abstract

The Cosmic Neutrino Background (CνB) is the relic background of neutrinos from the early universe, predicted to have a temperature $T_\nu = (4/11)^{1/3} T_\gamma \approx 1.946$ K and a number density $n_\nu \approx 336$ cm^{-3} per flavor. In Recognition Science, neutrinos are fermionic (4-tick, half-integer spin) ledger modes carrying the three Dirac neutrino masses at deep rungs $r_\nu \in \{-54, -58, -63\}$ on the φ-ladder (Lean: `Physics.NeutrinoSector`, proved). The CνB temperature follows from the RS-derived CMB temperature $T_0 = 2.725$ K (`RS_CMB_Temperature.tex`) combined with the $(4/11)^{1/3}$ cooling factor from e^+e^- annihilation. The neutrino number density uses the Fermi-Dirac statistics for massless (or ultra-relativistic) fermions (Lean: `Thermodynamics.FermiDirac`). The CνB contributes $N_{\text{eff}} = 3.046$ effective neutrino species to the radiation density.

1 Introduction

The CνB is the neutrino analogue of the CMB. Neutrinos decouple at $T \sim 2$ MeV (weak interaction freeze-out), then e^+e^- annihilation heats the photon bath but not the neutrinos, giving $T_\nu/T_\gamma = (4/11)^{1/3}$.

2 RS Framework

2.1 Neutrino Masses from RS

Theorem 2.1 (Neutrino masses, Lean-proved)**.** *The three neutrino masses occupy rungs $r_{\nu_3} = -54$, $r_{\nu_2} = -58$, $r_{\nu_1} = -63$ (normal hierarchy) on the Dirac φ-ladder:*

$$m_{\nu_3} \approx \sqrt{\Delta m^2_{32}} \approx 0.050\, eV, \tag{1}$$

$$m_{\nu_2} \approx \sqrt{\Delta m^2_{21}} \approx 0.0086\, eV. \tag{2}$$

Lean: `Physics.NeutrinoSector.nu3_frac_pred_bounds`, `Physics.NeutrinoSector.dm2_21_frac_pred_in_nufit_1sigma`

2.2 Fermionic Statistics for Neutrinos

Neutrinos are spin-1/2 fermions (4-tick ledger modes):

$$\bar{n}_\nu(\varepsilon) = \frac{1}{\exp(\varepsilon/T_\nu) + 1} \quad (m_\nu \ll T_\nu \text{ for C}\nu\text{B}). \tag{3}$$

`Lean: Thermodynamics.FermiDirac`
`Lean: Foundation.EightTick.phase_4_is_minus_one (fermion)`

3 Derivation of CνB Temperature

3.1 Decoupling Temperature

Neutrinos decouple when $\Gamma_{\text{weak}} \sim H$:

$$T_{\text{dec}} \approx \left(\frac{m_W^4}{M_{\text{Pl}}}\right)^{1/3} \approx 2\,\text{MeV}. \tag{4}$$

With $m_W \approx 80.4$ GeV (RS: `predict_mass`) and $M_{\text{Pl}} = c^2\sqrt{\hbar c/G}$ (RS: `G_rs_eq`): $T_{\text{dec}} \approx 2$ MeV.

3.2 e^+e^- Annihilation Heating

After decoupling, e^+e^- annihilation heats photons:

$$T_\nu = \left(\frac{4}{11}\right)^{1/3} T_\gamma \approx 0.7138\, T_\gamma. \tag{5}$$

Theorem 3.1 (CνB temperature)**.** *With $T_\gamma = T_0 = 2.72548\ K$ (Lean: `Physics.CMB.rs_cmb_approx_27`*

$$T_\nu = \left(\frac{4}{11}\right)^{1/3} \times 2.72548\, K = 1.9454\, K. \tag{6}$$

*Experimental target: $T_\nu = 1.9454\ K$ (PTOLEMY experiment [**?**]).*

3.3 Number Density

The CνB number density per flavor:

$$n_\nu = \frac{3}{11} n_\gamma = \frac{3}{11}\frac{2\zeta(3)}{\pi^2} T_\gamma^3 \approx 56\,\text{cm}^{-3} \text{ per flavor} \times 2\,(\nu + \bar{\nu}) = 112\,\text{cm}^{-3}. \tag{7}$$

Total (3 flavors): $n_{\nu,\text{total}} \approx 336\ \text{cm}^{-3}$.

3.4 Effective Relativistic Degrees of Freedom

The CνB contributes to the radiation density:

$$\rho_{\text{rad}} = \rho_\gamma \left[1 + \frac{7}{8} N_{\text{eff}} \left(\frac{T_\nu}{T_\gamma}\right)^4\right], \quad N_{\text{eff}} = 3.046. \tag{8}$$

The $N_{\text{eff}} = 3.046$ (slightly above 3) arises from incomplete neutrino decoupling and QED corrections. In RS, this correction uses the finite-temperature QED structure (`RS_Electroweak_Phase_Transition.tex`).

4 Numerical Results

Quantity	RS Prediction	Standard Value	Status
T_ν	1.9454 K	1.9454 K	✓
n_ν (per flavor)	112 cm^{-3}	112 cm^{-3}	✓
N_{eff}	3.046 (SM)	3.046 (Planck)	✓
ν masses (normal hierarchy)	Rung $r \in \{-54, -58, -63\}$	NuFIT 5.3	Within 1σ
T_{dec}	≈ 2 MeV	≈ 2 MeV	✓

5 Lean Formalization Status

Claim	Lean theorem	Module	St
Neutrino masses (rungs)	`nu3_frac_pred_bounds`	`Physics.NeutrinoSector`	P
Δm^2_{21} in NuFIT 1σ	`dm2_21_frac_pred_in_nufit_1sigma`	`Physics.NeutrinoSector`	P
Fermi-Dirac ν statistics	`FermiDirac`	`Thermodynamics`	P
Neutrino spin-1/2 (fermionic)	`phase_4_is_minus_one`	`Foundation.EightTick`	P
CMB temperature T_γ	`rs_cmb_approx_2725`	`Physics.CMBTemperature`	P
W boson mass	`predict_mass`	`Masses.MassLaw`	P
$(4/11)^{1/3}$ cooling	—	—	H
$N_{\text{eff}} = 3.046$ correction	—	—	H

† **HYPOTHESIS**: $(4/11)^{1/3}$ factor from e^+e^- annihilation in RS thermodynamics. Falsifier: direct CνB detection (PTOLEMY) showing $T_\nu \neq 1.946$ K.

‡ **HYPOTHESIS**: $N_{\text{eff}} = 3.046$ from RS finite-temperature QED. Falsifier: CMB-S4 measurement of N_{eff} deviating from 3.046 by $> 3\sigma$.

6 Conclusion

The cosmic neutrino background temperature ($T_\nu = 1.946$ K), number density ($n_\nu = 336$ cm^{-3}), and neutrino mass spectrum are all derived from RS: neutrino masses from the φ-ladder (Lean-proved), CMB temperature from the RS recombination calculation (Lean-proved), and fermionic statistics from the 8-tick structure (Lean-proved).

References

[1] E. W. Kolb, M. S. Turner, *The Early Universe* (Addison-Wesley, 1990).

[2] PTOLEMY Collaboration, "Neutrino physics with the PTOLEMY project," *J. Cosmology Astropart. Phys.* **2022**, 036 (2022).

[3] J. Washburn, "The Algebra of Reality," *Axioms* **15**(2), 90 (2026).